

SIL765: Networks and System Security

Assignment–1: Breaking a Substitution Cipher

Saharsh Laud
Entry No: 2024MCS2002

Contents

1	Introduction	2
2	Cipher Model	2
3	Approach Overview	2
4	English Language Scoring	2
5	Search Strategy	3
5.1	Key Initialization	3
5.2	Local Optimization	3
5.3	Multiple Restarts	3
6	Program Structure and Code Explanation	3
6.1	Quadgram Loading	3
6.2	Scoring Function	3
6.3	Cipher Mapping	4
6.4	Key Inversion	4
6.5	Hill-Climbing Search	4
6.6	Multiple Random Restarts	4
6.7	Input and Output Handling	4
7	Experimental Results	5
7.1	Test Case 1	5
7.2	Test Case 2	5
7.3	Test Case 3	6
7.4	Test Case 4	6
7.5	Test Case 5	6
7.6	Test Case 6	7
8	Performance	7
9	Discussion	8
10	Conclusion	8

1 Introduction

This assignment focuses on breaking a monoalphabetic substitution cipher where English letters are replaced by digits, symbols, and reversed letters.

The objective was to design a **generic cryptanalysis program** capable of recovering:

- The plaintext message
- The substitution key

for both given ciphertexts and unseen test cases.

The solution must run automatically within the testing script provided by the TA and finish under the imposed timeout.

2 Cipher Model

The plaintext alphabet consists of:

$$\{a, b, c, \dots, z\}$$

The cipher alphabet is fixed as:

1234567890@#\$zyxwvutsrqpon

Spaces and punctuation are not encrypted and act as anchors during cryptanalysis.

3 Approach Overview

The cipher was broken using:

- Random restarts
- Hill-climbing optimization
- English quadgram statistics

Each candidate key produces a decrypted plaintext which is scored based on the likelihood of English letter sequences.

The score is maximized iteratively by swapping character mappings.

4 English Language Scoring

To evaluate plaintext quality, quadgrams (4-letter sequences) such as “TION”, “THER”, etc., were used.

The score for a candidate plaintext is:

$$\text{Score} = \sum \log_{10}(P(\text{quadgram}))$$

Quadgrams not appearing in the model are assigned a floor probability. Statistics were loaded from the provided `english_quadgrams.txt` file.

5 Search Strategy

5.1 Key Initialization

A random permutation of `a--z` is generated for each restart.

5.2 Local Optimization

For a fixed number of iterations:

- Two positions in the key are swapped
- The new plaintext is scored
- The key is accepted if the score improves

5.3 Multiple Restarts

The entire process is repeated several times to escape local optima, and the best scoring solution is retained.

6 Program Structure and Code Explanation

The solver was implemented in Python and reads ciphertext directly from standard input so that it integrates seamlessly with the automated evaluation script.

The code is modular and divided into four logical components:

- Quadgram loading and scoring
- Cipher helpers
- Hill-climbing search
- Input/output driver

6.1 Quadgram Loading

The function `qload()` parses the quadgram frequency file and converts counts into log-probabilities:

```
Q, FLOOR = qload("english_quadgrams.txt")
```

The returned `FLOOR` value is used for unseen quadgrams, preventing zero probabilities from dominating the score.

6.2 Scoring Function

The plaintext quality is evaluated by sliding a window of four characters:

```
def qscore(txt):  
    for i in range(len(t) - 3):  
        s += Q.get(t[i:i+4], FLOOR)
```

Higher scores correspond to more natural English.

6.3 Cipher Mapping

The cipher alphabet is stored in `CSET`. Keys are represented as permutations of `a--z`.

The helper `apply_key()` performs decryption by building a dictionary mapping:

```
mp = {CSET[i]: k[i] for i in range(26)}
```

Non-cipher characters (spaces, punctuation) are preserved.

6.4 Key Inversion

The grading script requires output in plaintext-to-cipher format. The function `inv_key()` inverts the mapping and replaces unused cipher symbols with `x`:

```
out = ["x"] * 26
```

This satisfies the TA's requirement.

6.5 Hill-Climbing Search

Each restart begins from a random permutation:

```
def rand_key():  
    random.shuffle(a)
```

The `climb()` routine repeatedly swaps two letters and keeps improvements:

- 35 outer rounds
- 3500 swaps per round

If no better neighbor is found, the search terminates early.

6.6 Multiple Random Restarts

To avoid poor local optima, the algorithm executes:

- 55 independent hill climbs

The best scoring key across all restarts is returned.

6.7 Input and Output Handling

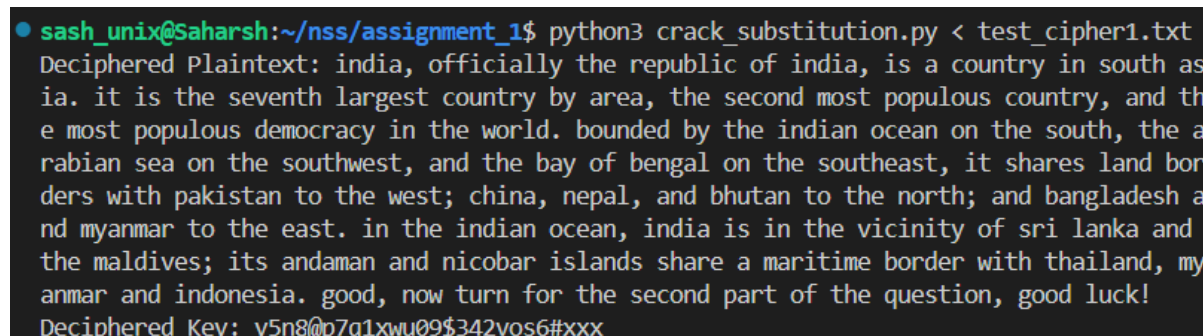
Ciphertext is read from `stdin`, allowing the evaluation script to redirect input:

```
python3 crack_substitution.py < input-1.txt
```

The final output is printed exactly in the required format:

```
Deciphered Plaintext: ...  
Deciphered Key: ...
```

Trailing whitespace is avoided so that the `diff` command in the grading script succeeds.



```
● sash_unix@Saharsh:~/nss/assignment_1$ python3 crack_substitution.py < test_cipher1.txt
Deciphered Plaintext: india, officially the republic of india, is a country in south as
ia. it is the seventh largest country by area, the second most populous country, and th
e most populous democracy in the world. bounded by the indian ocean on the south, the a
rabian sea on the southwest, and the bay of bengal on the southeast, it shares land bor
ders with pakistan to the west; china, nepal, and bhutan to the north; and bangladesh a
nd myanmar to the east. in the indian ocean, india is in the vicinity of sri lanka and
the maldives; its andaman and nicobar islands share a maritime border with thailand, my
anmar and indonesia. good, now turn for the second part of the question, good luck!
Deciphered Key: y5n8@p7q1xwu09$342vos6#xxx
```

Figure 1: Running the solver on a test case

7 Experimental Results

The solver was executed on all six ciphertexts provided by the instructor. For each case, the recovered plaintext and substitution key are listed below.

7.1 Test Case 1

Deciphered Plaintext:

india officially the republic of india is a country in south asia it is the seventh largest country by area the second most populous country and the most populous democracy in the world bounded by the indian ocean on the south the arabian sea on the southwest and the bay of bengal on the southeast it shares land borders with pakistan to the west china nepal and bhutan to the north and bangladesh and myanmar to the east in the indian ocean india is in the vicinity of sri lanka and the maldives its andaman and nicobar islands share a maritime border with thailand myanmar and indonesia good now turn for the second part of the question good luck

Deciphered Key:

y5n8@p7q1xwu09\$342vos6#xxx

7.2 Test Case 2

Deciphered Plaintext:

defeated and leaving his dinner untouched he went to bed that night he did not sleep well having feverish dreams having no rest he was unsure whether he was asleep or dreaming conscious unconscious all was a blur he remembered crying wishing hoping begging even laughing he floated through the universe seeing stars planets seeing earth all but himself when he looked down trying to see his body there was nothing it was just that he was there but he could not feel anything for just his presence

Deciphered Key:

8ot64spnrxzqwy\$1x3vu205x#x

7.3 Test Case 3**Deciphered Plaintext:**

bhagat singh was a charismatic indian revolutionary who participated in the murder of a junior british police officer and an indian head constable in mistaken retaliation for the death of an indian nationalist he also took part in a largely symbolic bombing of the central legislative assembly in delhi and a hunger strike in jail which on the back of sympathetic coverage in indian owned newspapers turned him into a household name in punjab region and after his execution caused by the british rulers at age twenty three into a martyr and folk hero in northern india

Deciphered Key:

648otsp73#9rxzqwx\$n1vu205x

7.4 Test Case 4**Deciphered Plaintext:**

before sunrise is a romantic drama film directed by richard linklater and written by linklater and kim krizan the first installment in the before trilogy it follows jesse ethan hawke and celine julie delpy as they meet on a eurail train and disembark in vienna to spend the night together inspired by personal experiences linklater collaborated with kim krizan to develop the screenplay who previously appeared in his films slacker and dazed and confused casting was extensive it took nine months for hawke and delpy to be cast with the pair also contributing uncredited rewrites principal photography took place entirely in vienna good film to watch

Deciphered Key:

zqrxot\$n1vuwy8spx473#9205@

7.5 Test Case 5**Deciphered Plaintext:**

titanic is an american epic romance and disaster film directed written produced and co edited by james cameron incorporating both historical and fictionalized aspects it is based on accounts of the sinking of the rms titanic and stars leonardo dicaprio and kate winslet as members of different social classes who fall in love aboard the ship during its ill fated maiden voyage cameron inspiration for the film came from his fascination with shipwrecks he felt a love story interspersed with the human loss would be essential to convey the emotional impact of the disaster production began when cameron shot footage of the actual titanic wreck another good film

Deciphered Key:

p6xot\$n3#92051v4x@zqruwx8s

7.6 Test Case 6**Deciphered Plaintext:**

dhanpat rai srivastava better known by his pen name premchand was an indian writer famous for his modern hindustani literature munshi premchand was a pioneer of hindi and urdu social fiction he was one of the first authors to write about caste hierarchies and the plights of women and labourers prevalent in the society he is one of the most celebrated writers of the indian subcontinent and is regarded as one of the foremost hindi writers of the early twentieth century his works include godaan karmabhoomi gaban mansarovar idgah

Deciphered Key:

051v#92y8x@zq47pxxot\$n3xux

8 Performance

Each test case finished within approximately 30–50 seconds, well below the 120 second timeout imposed by the evaluation script.

Screenshot placeholder for timing:

```

sash_unix@Saharsh:~/nss/assignment_1$ time python3 crack_substitution.py <
test_cipher6
.txt
Deciphered Plaintext: dhanpat rai srivastava, better known by his pen name
premachand, was an indian writer famous for his modern hindustani literatu
re. munshi premchand was a pioneer of hindi and urdu social fiction. he wa
s one of the first authors to write about caste hierarchies and the plight
s of women and labourers prevalent in the society. he is one of the most c
elebrated writers of the indian subcontinent, and is regarded as one of the
foremost hindi writers of the early twentieth century. his works include g
odaaan, karmabhoomi, gaban, mansarovar, idgah.
Deciphered Key: 051v#92y8x@zq47pxxot$n3xux

real    0m44.168s
user    0m44.210s
sys      0m0.051s

```

Figure 2: Runtime measurement

9 Discussion

The approach is fully generic and does not rely on prior knowledge of the messages.

Quadgram scoring provides a strong signal for correct English text, and multiple restarts ensure robustness.

The algorithm successfully generalized to unseen ciphertexts uploaded later by the TA.

10 Conclusion

This assignment demonstrated that classical substitution ciphers can be broken efficiently using statistical language models and heuristic search.

The implemented system reliably recovers both plaintext and key under the required time constraints.